

Makaziwe Lubisi

Exercise 5 - NULL Functions

Q1

```
SELECT employee-id,  
       name,  
       IFNULL (salary, 0) AS salary-with-default
```

employee-id	name	salary-with-default
1	Alice	8000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2

```
SELECT employee-id,  
       name,  
       IFNULL (department, 'Not Assigned') AS  
       department_name  
FROM employees;
```

employee-id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Q3

```
SELECT order_id,  
       customer_id  
FROM orders  
WHERE customer_id IS NULL ;
```

order_id	customer_id
103	NULL

Q4.

```
SELECT order_id,  
       customer_id,  
       IFNULL (delivery_date, 'Pending') AS delivery_status  
FROM orders ;
```

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5

```
SELECT student_id,  
       name,  
       IFNULL (grade, 0) AS final_grade  
FROM students ;
```

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6

```
SELECT COUNT(*) AS not_graded_count
FROM students
WHERE grade IS NULL ;
```

not_graded_count
1

Q7

```
SELECT product_id, name, price
IFNULL (discount, 0) AS final_price
FROM products ;
```

product_id	name	final_price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Q8.

```
SELECT COUNT(*) AS missing_email_count
FROM customers
WHERE email IS NULL ;
```

missing_email_count
2

Q9

```
SELECT customer_id,  
       name,  
       IFNULL(email, 'No email') AS email_display  
FROM customers ;
```

customer_id	name	email_display
1	Linda	No email
2	Joseph	Joseph@mail.com
3	Nia	No email.

Q10

```
SELECT payment_id, method, status,  
       status  
       IFNULL(method, 'Unknown') AS method_display  
FROM payments ;
```

payment_id	method_display	status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed.

Q11

```
SELECT item_id, item_name  
       IFNULL(quantity, 0) AS quantity_checked  
FROM inventory ;
```

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Q12

```
SELECT emp_id,  
       COALESCE (bonus, commission, 0) AS first_available_reward  
FROM Employees - Extra ;
```

emp_id	first_available_reward
1	300
2	100
3	0

Q13

```
SELECT COUNT(*) AS no_room_count  
FROM classes  
WHERE room IS NULL ;
```

no_room_count

2

Q14

```
SELECT student_id,  
       date,  
       IFNULL (status 'Not marked') AS attendance_status
```

student_id	date	attendance_status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

Q15

```
SELECT account-id,  
IFNULL (account-type, 'Unknown') AS type-display,  
IFNULL (balance, 0) AS balance-checked  
FROM Bank-Account ;
```

account-id	type-display	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q16

```
SELECT project-id,  
title  
IFNULL (start-date, 'TBD') AS start-display  
FROM projects ;
```

project-id	title	start-display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data migration	TBD

Q17

```
SELECT review-id, product-id  
IFNULL (comment, 'No comment') AS comment-display  
IFNULL (rating, 0) AS rating-display  
FROM reviews
```

review-id	product-id	comment-display	rating-display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3.

Q18.

```
SELECT  supplier-id,
        name ,
        COALESCE (phone, alt-phone, 'No contact')
        AS contact-number
FROM    Suppliers
```

Supplier-id	name	contact-number
1	Gibbal Goods	123456789
2	Best Supplies	987654321
3	Value Source	No contact.

Q19

```
SELECT  user-id,
        IFNULL (theme, 'light') AS theme-set
        IFNULL (language, 'English') AS language-set
        IFNULL (timezone, 'UTC') AS timezone-set
FROM    user-settings ;
```

user-id	theme-set	language-set	timezone-set
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Q20

```
SELECT record-id,  
       machine-id,  
       IFNULL(issue, 'Unknown Issue') AS issue-log  
       IFNULL(technician, 'Not Assigned') AS technician-name  
FROM maintenance;
```

record-id	machine-id	issue-log	technician-name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex